

Sovereign On-Premise Agentic AI Workbench

Using Open-Weight Multimodal LLMs for Confidential Industrial Work

SIH 2026 – Problem Statement Analysis

A structured analysis of pain points, feasibility, innovation, impact, architecture, competition, risks, and evaluator strategy.

GREEN LIGHT

Strong problem relevance with a technically demonstrable sovereignty and air-gap advantage.

Core Principle:

Sovereignty is the story. The model is the engine.

August 26, 2026

Executive Summary

The proposed solution is a sovereign, on-premise, agentic AI workbench designed for organizations that cannot send confidential industrial data to external AI APIs.

The platform combines:

- Open-weight multimodal large language models.
- Fully on-premise inference.
- Agentic orchestration.
- Multimodal document and image understanding.
- Retrieval-Augmented Generation (RAG).
- Enterprise authentication and role-based access control.
- Compliance-oriented audit logging.
- Network isolation and zero-egress verification.
- Deployable infrastructure suitable for air-gapped environments.

GREEN LIGHT

The strongest differentiator is not simply running an LLM locally. The proposed system turns local inference into a verifiable, auditable and agentic enterprise workbench.

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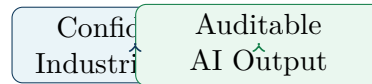
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1 Pain Points and Core Understanding

1.1 The Exact Problem

Industries handling defense blueprints, pharmaceutical batch records, oil and gas SCADA logs, semiconductor process data, or citizen PII may be unable or unwilling to send sensitive information to cloud-based AI providers.

The proposed workbench addresses this requirement by providing:



1.2 Root Causes

Root Cause	Why It Persists
Regulatory requirements	Organizations operate under frameworks such as IEC 62443, 21 CFR Part 11, EU AI Act, CMMC and NERC CIP. Data handling, auditability and residency requirements can restrict cloud usage.
IP sensitivity	Industrial process parameters, formulations, semiconductor designs, engineering documents and operational data may represent highly valuable intellectual property.
Frontier model lock-in	Cloud frontier models cannot simply cross a true physical or logical air gap.
Lack of turnkey sovereign stacks	Organizations often need to assemble inference, RAG, authentication, agent orchestration, monitoring and security components manually.
Multimodal gap	Industrial workflows frequently contain PDFs, diagrams, scanned SOPs, instrument images, tables and technical drawings rather than plain text.

1.3 Primary Stakeholders

- Manufacturing plant managers.
- Defense and aerospace engineering teams.
- Pharmaceutical QA teams.
- Critical infrastructure operators.
- Public-sector organizations and PSUs.
- Industrial R&D laboratories.
- Sovereign data and government technology missions.

1.4 Current Inefficiencies

- Significant engineering effort is spent on infrastructure plumbing.
- Offline multimodal ingestion remains difficult.
- Air-gap verification is rarely integrated into the application itself.
- Agent orchestration, RAG and authentication are often separate projects.
- There is no universally accepted “one-command” sovereign AI workbench.

2 Feasibility of Execution

2.1 What Can Be Built in 36–48 Hours

Layer	Technology	Reason
Inference	Ollama → vLLM	Ollama provides developer-friendly local inference; vLLM is suitable for production-style serving.
Base Model	Qwen2.5-VL-7B or Llama-3.2-11B-Vision	Open-weight multimodal models suitable for a demonstrable prototype.
Agent	LangGraph	Deterministic state-machine-style agent orchestration.
Prototyping	CrewAI	Useful for rapidly demonstrating role-based multi-agent workflows.
Vector Database	Qdrant or Chroma	Local RAG without requiring external SaaS infrastructure.
UI	Open WebUI + React dashboard	Fast route to a usable local AI interface.
Authentication	Keycloak / Authentik	Demonstrates enterprise identity and role management.
Sandbox	Network namespace + firewall	Provides a concrete mechanism for demonstrating zero-egress operation.

2.2 Likely Blockers

Risk / Limitation

- GPU memory constraints.
- Lack of genuine industrial confidential data.
- Hidden telemetry or network dependencies.
- Container and model download dependencies.
- Cold-start latency for large models.
- Agent loops and hallucinated tool calls.
- Complexity of compliance evidence.

2.3 Recommended MVP

Key Insight

SOP-QA Copilot for an anonymized pharmaceutical production line

The user uploads a scanned SOP PDF and an image of an instrument panel. The system extracts relevant information, reasons over the documents, checks compliance-oriented rules, identifies deviations and generates an audit-ready report.

The demonstration concludes with a live zero-egress verification.

3 Impact and Relevance

3.1 Beneficiary Map

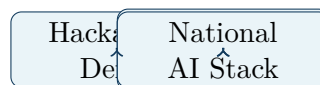
Beneficiary	Concrete Use Case
Industrial MSMEs	Affordable private AI without mandatory cloud inference.
Defense and aerospace organizations	Confidential technical-document analysis.
Pharmaceutical companies	SOP analysis, deviation triage and controlled document assistance.
Critical infrastructure	SCADA-log summarization and operational assistance.
Research laboratories	Private AI experimentation and sovereign AI research.
Government data missions	On-premise AI for sensitive government work-flows.

3.2 Strategic Impact

The platform aligns with major themes around:

- Data sovereignty.
- Indigenous AI infrastructure.
- Confidential industrial computing.
- Responsible and auditable AI.
- Secure deployment of open-weight models.
- IndiaAI and Atmanirbhar Bharat objectives.

3.3 Scalability Ladder



4 Scope of Innovation

4.1 Competitor Landscape

Player / Project	Strengths	Gaps
Open WebUI	Polished local chat interface; Ollama integration	Primarily chat-centric; additional agent infrastructure required.
Enterprise Air-Gapped LLM	Reference architecture for enterprise deployment	Limited agentic and multimodal workbench capabilities.
Zorost Air-Gapped Stack	Manufacturing and compliance orientation	Vendor-oriented rather than a fully general deployable workbench.

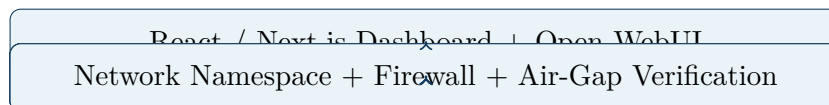
TrueFoundry Air-Gapped Tier	Strong enterprise deployment and compliance	Commercial enterprise offering.
QID-NPI	Edge-native architecture for specialized environments	Niche use case; not a general industrial AI workbench.
Closed SaaS AI Platforms	Excellent UX and mature models	Cloud dependency conflicts with strict sovereignty requirements.
NVIDIA NIM / NeMo	Strong inference optimization	Not fully aligned with a completely open-weight, vendor-neutral stack.

4.2 Innovation Differentiators

Differentiator	Value / Wow Factor
Air-gap verification panel	Live network telemetry and DNS-leak monitoring.
Model hot-swap orchestrator	Switch between Qwen, Llama, Mistral or other supported models.
Multimodal SOP-RAG	Process text, diagrams, tables and images from a single document.
Compliance-as-code	Generate structured audit evidence and compliance-oriented logs.
No-code agent builder	Convert tool graphs into deployable agent configurations.
Signed model provenance	Verify model artifacts using cryptographic hashes.
Per-tenant RBAC	Separate departments, roles and data namespaces.

5 Reference Architecture

5.1 High-Level Architecture



5.2 Recommended Technology Stack

- **Model:** Qwen2.5-VL or Llama Vision.
- **Inference:** Ollama for development; vLLM for production.
- **Agent orchestration:** LangGraph.
- **Vector store:** Qdrant.
- **Frontend:** Next.js / React.
- **Authentication:** Keycloak or Authentik.
- **Observability:** Prometheus + Grafana.

- **Isolation:** Docker/network namespaces + firewall rules.
- **Model provenance:** SHA-256 or equivalent cryptographic verification.

6 Clarity of the Problem Statement

The platform must provide four fundamental capabilities:

1. **Sovereignty:** all inference and confidential data remain on-premise.
2. **Multimodality:** text, images, scanned documents, tables and diagrams.
3. **Agentic capability:** planning, tool use, RAG, memory and multi-step execution.
4. **Industrial context:** SOPs, blueprints, batch records, telemetry and technical documents.

6.1 Common Misinterpretations

Critical Red Flag

Avoid presenting the project as “just another RAG chatbot.”
A strong solution must demonstrate:

- Agent orchestration.
- Tool execution.
- Multimodal ingestion.
- Auditability.
- Authentication.
- Zero-egress verification.

7 Air-Gap and Sovereignty Strategy

7.1 Three-Layer Proof

The strongest demonstration should provide three independent forms of evidence.

1. **Firewall evidence:** outbound network policy set to deny.
2. **Live network evidence:** packet capture showing no unexpected outbound traffic.
3. **Artifact evidence:** model manifests and hashes verified locally.

Example Linux demonstration:

```
sudo iptables -P OUTPUT DROP
sudo iptables -L OUTPUT -v -n

sudo tcpdump -i any -nn
```

Offline environment variables can include:

```
export TRANSFORMERS_OFFLINE=1
export HF_HUB_OFFLINE=1
```


Risk / Limitation

A genuine air-gap claim must be demonstrated, not merely stated.

All model weights, container images, dependencies, tokenizers, embedding models and required certificates must be available locally before network isolation is enabled.

8 Evaluator Perspective

8.1 Likely Evaluation Dimensions

Criterion	Estimated Weight	What Wins
Feasibility	25%	Reliable end-to-end live demo.
Innovation	20%	Air-gap verifier + multimodal agentic workflow.
Technical Depth	20%	Inference, RAG, agents, security and evaluation architecture.
Impact	15%	Industrial, national and sovereignty relevance.
Completeness	10%	Polished UI, deployment script and documentation.
Presentation	10%	Clear pitch and confident technical Q&A.

8.2 Red Flags

- Fake “sovereignty” while using cloud APIs.
- No live demonstration.
- Single-user chatbot presented as an agentic platform.
- Unsupported compliance claims.
- Claiming hallucination-free AI.
- No measurable evaluation methodology.
- No network isolation proof.

9 Strategy for Team Fit and Execution

9.1 Recommended Skill Mix

Role	Responsibility
Backend / MLOps	Inference, APIs, Docker, vector database and deployment.
AI / ML Engineer	Prompting, RAG, agents, evaluation and model integration.

Frontend / UX	Dashboard, document workflows, visualization and user experience.
DevSecOps	Network isolation, authentication, secrets, model provenance and security.
Domain / Pitch Lead	Industrial requirements, compliance research and presentation.

9.2 Execution Timeline

Time	Task
Day -7 to -5	Research inference, agent and air-gap architectures.
Day -5 to -3	Select multimodal model and benchmark it on target hardware.
Day -3 to -1	Build Planner → RAG → Tool LangGraph workflow.
Day -1	Rehearse network isolation and zero-egress demonstration.
Day 0	Freeze architecture, polish UI and prepare backup video.

10 AI-Buildability Split

10.1 What AI Can Build Quickly

- FastAPI boilerplate.
- React / Next.js scaffolding.
- LangGraph starter graphs.
- Docker Compose configurations.
- Ollama / vLLM launch scripts.
- Documentation.
- Synthetic SOP generation.
- Unit tests and CI templates.

10.2 What Requires Real Engineering

Concern	Why It Is Difficult
Verifiable air gap	Telemetry and hidden network dependencies may exist inside dependencies.
Multimodal chunking	Industrial documents contain diagrams, tables, scans and structured information.
Agent reliability	Agents can loop, hallucinate tool calls or produce incorrect actions.
Compliance evidence	Audit requirements are domain-specific and must be verified.
GPU memory	Vision models, context and RAG can exceed available VRAM.

Evaluation

Offline evaluation requires carefully constructed datasets and metrics.

Critical Red Flag

AI-generated infrastructure and compliance claims must be manually verified. Never assume that generated code or regulatory statements are correct merely because they compile or sound authoritative.

11 Data and Resource Availability

11.1 Public Data Sources

- PubMed Central Open Access.
- Hugging Face datasets and model repositories.
- Manufacturing and industrial QA datasets.
- OSHA publications.
- Public standards and compliance documentation where legally available.

11.2 Synthetic Data Strategy

Because confidential DRDO, HAL, pharmaceutical or semiconductor data will generally not be available to a hackathon team, construct a synthetic dataset.

Recommended structure:

```
SOP-001
|- Assembly procedure
|- Calibration procedure
|- Deviation procedure
|- Equipment image
|- Expected QA pairs

SOP-002
|- ...
```

Target:

100 synthetic SOPs $\times$ 5 annotated images per SOP

This produces a controlled evaluation environment while avoiding the use of confidential information.

12 Judge Q&A Stress Test

12.1 Q1. How is this different from simply running Ollama?

Key Insight

Ollama is the inference engine. The proposed system is the workbench. The workbench adds agent orchestration, multimodal RAG, RBAC, audit logging, network isolation, provenance verification and enterprise deployment.

12.2 Q2. Prove that no data leaves the machine.

Answer:

We provide three independent layers of evidence:

1. Firewall policy showing outbound traffic is denied.
2. Live packet capture showing no unexpected external traffic.
3. Cryptographic verification of locally stored model artifacts.

12.3 Q3. Why not simply use GPT or Claude?

The strongest answer is not “open models are always better.”

Instead:

- Some environments require strict data sovereignty.
- Some environments are physically air-gapped.
- Confidential IP may not be permitted to leave the network.
- Local inference provides predictable infrastructure ownership.
- Open-weight models provide model portability and deployment control.

12.4 Q4. What is your hallucination rate?

Do not invent a number.

Use an offline evaluation dataset and report:

- Faithfulness.
- Answer relevance.
- Citation correctness.
- Unsupported-claim rate.
- Tool execution success rate.

12.5 Q5. Can this scale to hundreds of concurrent users?

The prototype should not pretend that a single workstation can serve hundreds of users.

Instead:

Ollama prototype → vLLM production serving → GPU scaling → horizontal inference replicas

This demonstrates architectural scalability without making unsupported performance claims.

13 Recommended MVP

GREEN LIGHT

Bharat-Workbench

A one-command deployable sovereign agentic AI platform that:

1. Ingests confidential industrial PDFs and images.
2. Extracts multimodal information.
3. Retrieves relevant SOP knowledge.
4. Uses an open-weight multimodal LLM.
5. Executes controlled tools through LangGraph.
6. Produces citations and evidence.
7. Generates audit-oriented logs.
8. Enforces RBAC.
9. Verifies model provenance.
10. Demonstrates zero external network traffic.

14 Final Verdict

GREEN LIGHT

The problem is highly relevant and technically demonstrable, provided the team focuses on a narrow, reliable MVP rather than attempting to build an entire enterprise platform during the hackathon.

14.1 The Single Biggest Reason

The strongest story is not:

“We built another local LLM chatbot.”

It is:

“We built a sovereign industrial AI workbench that can prove, in real time, that confidential data never leaves the protected environment.”

The LLM is an implementation component.

The actual product differentiation is:

Sovereignty + Multimodality + Agents + RAG + Security + Auditability
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15 Final Summary Matrix

Dimension	Verdict	Assessment
Pain Points	GREEN	Real and strategically important.
Feasibility	GREEN	A focused MVP is achievable with existing open-source infrastructure.
Impact	GREEN	Strong relevance to sovereign and confidential AI deployment.
Innovation	AMBER	Individual components already exist; differentiation must come from integration and verifiable sovereignty.
Clarity	GREEN	Clear when framed as a sovereign workbench rather than a chatbot.
Evaluator Appeal	GREEN	Live air-gap verification provides a strong demonstration moment.
Team Fit	GREEN	Achievable with a 4–5 person technically balanced team.
AI Buildability	AMBER	Boilerplate is easy; security, reliability and compliance require real engineering.
Data	AMBER	Synthetic data will likely be required.
Q&A Risk	GREEN	Manageable if claims remain evidence-based.

16 Key References

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